Example 7.5 The length, breadth and height of a hall are 25 m, 15 m and 5 m respectively. Find the cost of renovating its floor and four walls at the rate of ₹80 per m^2.

Solution

Here, length (l) = 25 m, breadth (b) =15 m, height (h) = 5 m.

Area of four walls = LSA of cuboid

$= 2(l+b)\times h$

$= 2(25+15)\times 5$

$= 80\times 5 = 400\ m^2$

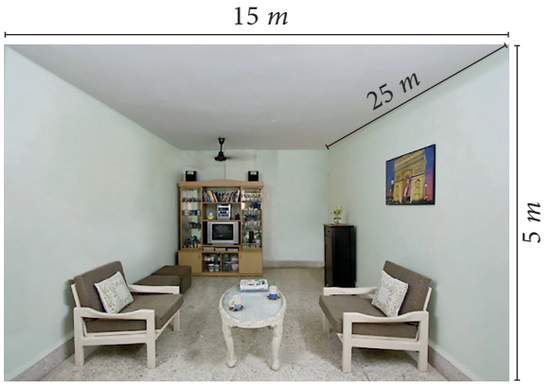


Fig. 7.12

Area of the floor $= l\times b$

$= 25\times 15$

$= 375\ m^2$

Total renovating area of the hall

= (Area of four walls + Area of the floor)

$= (400+375)\ m^2 \quad = 775\ m^2$

Therefore, cost of renovating at the rate of ₹80 per m^2 $= 80\times 775$

= ₹ 62,000

7.4.2 Cube and its Surface Area

Cube: A cuboid whose length, breadth and height are all equal is called as a cube. That is a cube is a solid having six square faces. Here are some real-life examples.


Dice


Ice cubes


Sugar cubes

Fig. 7.13

A cube being a cuboid has 6 faces, 12 edges and 8 vertices.

Consider a cube whose sides are 'a' units as shown in the Fig 7.14. Now,

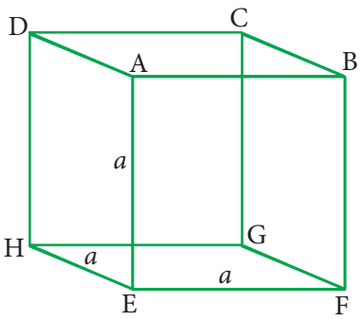


Fig. 7.14

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